Choose a motion method

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose a motion relationship. Explain what must be true for you to use it.

SUCCESS CRITERIA - I can...

1. Choose a relationship using the known quantities.
2. Show linked calculations with units and directions.
3. State when constant velocity or acceleration applies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

x_i=**-3±0.5 m**, v=**5±0.1 m/s** and exact t=**7 s**.
1. **Give central final position**, its conservative interval and full width.
2. **Name something you** are treating as true.

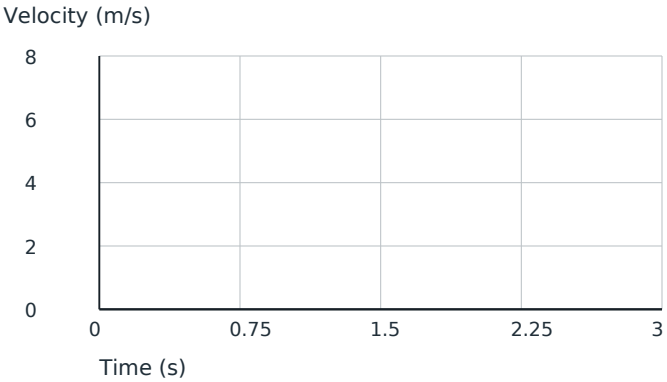
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $v_f = v_i + a\,\Delta t$; $\Delta x = v_i\,\Delta t + \frac{1}{2}a(\Delta t)^2$
$v_f^2 = v_i^2 + 2\,a\,\Delta x$.
Constant acceleration; i = initial, f = final. **Choose the equation containing your** knowns and unknown.

Worked reasoning: From rest, a cart accelerates uniformly at **2 m/s²** for **3 s**. Final velocity=**6 m/s**; average velocity=(**0**+**6**)/**2**=**3 m/s**; displacement=**3**×**3**=**9 m**.

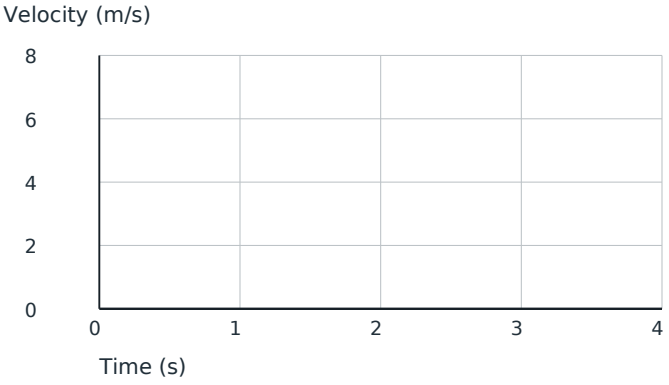
Your turn: Complete the cart's velocity-time graph. **Shade the triangular area under** the line and use it to **check the 9 m** displacement.



Marzano check - circle: ① ② ③

03 Solve and check

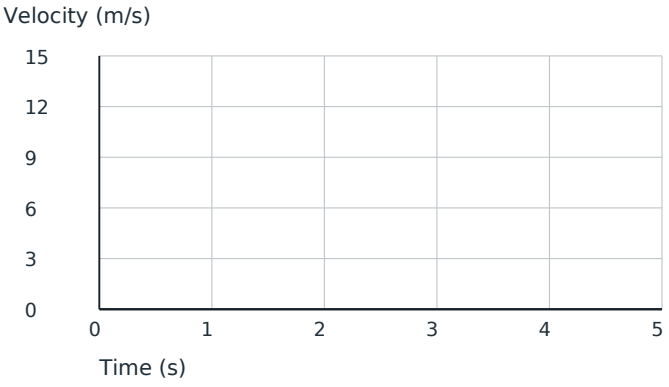
A cart starts at the **-3 m** mark on a test track, moving at **+2 m/s**. It accelerates at a constant **+1 m/s²** for **4 s**. **Draw its velocity-time graph**, then **find final velocity**, displacement and final position.



Marzano check - circle: ① ② ③

04 No time given

A cyclist brakes from **12 m/s** to rest with constant acceleration **-3 m/s²**. Forward is positive. **Find stopping distance** without first calculating time. Then **find the stopping time** and check with a velocity-time graph.



Marzano check - circle: ① ② ③

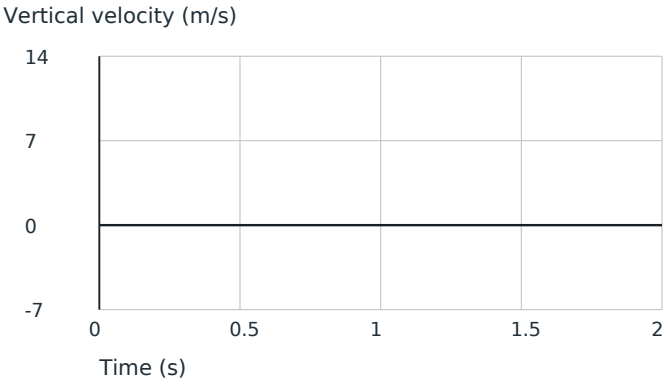
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Continue down each column.

05 At the top

You toss a ball straight upward at **14 m/s**. Ignore air resistance and use g = **9.8 m/s²**.
1. **Draw velocity** and acceleration arrows at release and at the highest point.
2. **Find the maximum rise** and time to the top.
3. Is acceleration zero there?



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Velocity changes from **-3** to **-15 m/s** in **3 s**, then stays at **-15 m/s** for **7 s**. **Find acceleration** and **describe the speed trend** in each interval.
Sketch the velocity–time graph; **label the given values**.



Marzano check - circle: ① ② ③

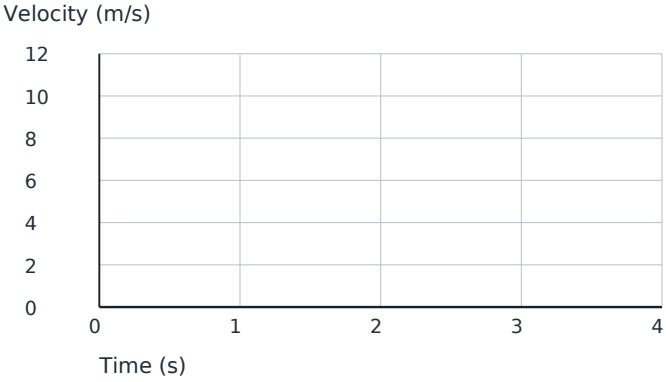
07 Exit check

A car starts at **+12 m/s** and brakes with a=**-3 m/s²** until it stops. A student extends the equation to **8 s** and reports v=**-12 m/s**. **Explain what the calculation treats** as true that no longer holds and **give stopping time**/distance.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A cart starts at **2 m/s** and travels **24 m** in **4 s** with uniform acceleration. Solve for acceleration in two ways and **explain why a constant-speed method** fails.



Marzano check - circle: ① ② ③